

MADSON ALLAN DE LUNA ARAGÃO

Bioinformatics Engineer, Software Development & Data Analyst

Belo Horizonte, MG, Brazil

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Portfolio | GitHub | LinkedIn | Google Scholar | ORCID

PROFESSIONAL SUMMARY

Bioinformatics Engineer and scientific software developer with approximately 10 years of research experience spanning genomics, multi-omics integration, experimental and computational transcriptomics, RNA-seq analysis, and the discovery and annotation of uncharacterized genes. Core expertise lies in molecular modeling and simulation, particularly classical molecular dynamics, peptide-membrane interactions, and peptide and protein design. Combines physics- and energy-based methods with deep learning for de novo and target-guided design. Develops pipelines, AI tools and applications for discovering, characterizing and optimizing candidates in omics datasets, integrating sequence, expression and structural evidence to support scientific decisions. Industry experience includes AI and computer vision to support malignant cell detection in clinical examinations and the development of automated hardware integrated with machine learning models. Holds a master's degree in Genetics and Molecular Biology from UFPE and is pursuing a PhD in Bioinformatics at UFMG. Currently Bioinformatics Engineer and Head of Data Science at Biotech Business and a member of the Nextflow Ambassadors program.

TECHNICAL SKILLS

- **Genomics and multi-omics integration:** genome mining, identification and annotation of uncharacterized genes, sequence analysis, and integration of genomic, transcriptomic and structural evidence for functional characterization and candidate prioritization.
- **Transcriptomics and RNA-seq:** experimental and computational experience, gene expression analysis and RNA-seq interpretation; integration of results to formulate hypotheses and support scientific decisions based on multi-omics data.
- **Molecular modeling and simulation:** protein, peptide and membrane modeling; docking, classical molecular dynamics, trajectory analysis, stability, flexibility and molecular interactions; GROMACS, HADDOCK and PyMOL.
- **Peptide and protein design:** de novo and target-guided design, sequence optimization and structural assessment; Rosetta for modeling and design, RFdiffusion for structure generation, ProteinMPNN for structure-conditioned sequence design, and RoseTTAFold and AlphaFold for structure prediction.
- **AI, computer vision and automation:** model development for pattern recognition and support for malignant cell detection in clinical images; integration of automated hardware, software and machine learning models; IoT system prototyping and validation.
- **Machine learning for omics data:** scikit-learn, Random Forest, SVM, gradient boosting, feature engineering and model comparison; deep learning, Transformers and generative models for gene, peptide and protein discovery, characterization and prioritization.
- **Programming and data analysis:** Python, R, Bash/Shell, SQL, pandas, NumPy, exploratory data analysis, applied statistics and data visualization; development of libraries and command-line tools.
- **Software development and data engineering:** FastAPI, Flask, REST APIs, GraphQL, React, JavaScript, HTML and CSS; PostgreSQL, MySQL, SQLite, Redis, API integration, ETL pipelines and data modeling.
- **Workflows, HPC and reproducibility:** Nextflow, Snakemake, Linux/Unix, Slurm, Docker and Docker Compose; automation of omics analyses and simulations, resource management, Git/GitHub, testing with pytest, documentation and modular code.
- **Product development and collaboration:** requirements gathering, prototyping, MVPs, delivery and backlog prioritization, product lifecycle tracking, roadmaps and communication with researchers, users and stakeholders; Scrum and Kanban.

PROFESSIONAL EXPERIENCE

Biotech Business

10/2025 - present

Bioinformatics Engineer and Head of Data Science

08/2026 - present

- **Requirements and priorities:** translate scientific and business needs into technical requirements and software deliverables, connecting data science activities with team needs.
- **Roadmaps and stakeholders:** contribute to roadmap definition and track solution development with research and business teams.
- **Computational solutions:** continue developing and integrating applications for organizing and analyzing biological data for agribusiness.

Mid-level Bioinformatician

10/2025 - 07/2026

- **Biological data and software:** developed computational solutions to organize, integrate and analyze information for agribusiness applications.
- **AI and computer vision:** developed applications using machine learning and image analysis to support analysis and research processes.
- **APIs and databases:** implemented services and integrations to make information accessible through team workflows and applications.

Seqera

01/2026 - present | Brazil | Hybrid

Nextflow Ambassador

- **Technical collaboration:** collaborate with Seqera developers in public GitHub repositories, contributing to the development and improvement of software products in the Nextflow ecosystem.
- **Testing and quality:** test tools and workflows across different use cases, identifying unexpected behavior, bugs and opportunities to improve the solutions.
- **Issues and evolution:** create, document and track GitHub issues with evidence and suggestions that support bug fixes and product evolution.
- **Code, tests and documentation:** contribute to improvements incorporated into Seqera repositories, including nf-metro: GitHub contribution.

PickCells

03/2020 - 08/2023 | Recife, PE, Brazil

Research and Development Analyst

08/2020 - 08/2023 | Recife, PE, Brazil | Hybrid

- **Developed AI and computer vision applications** for pattern recognition and support for malignant cell detection in hematology and cytology images, including cervical specimens and fluid samples from the pleuropulmonary region.
- **Contributed to the development of automated hardware** integrated with computer vision and machine learning models, connecting physical components with image analysis applications.
- **Prototyped, developed and validated** IoT, software and hardware systems, focusing on component integration and overall system operation.
- **Worked across the product lifecycle**, including requirements gathering, MVP development, incorporation of user feedback and improvement prioritization.
- **Collaborated with technical teams and stakeholders** to align user needs, implementation feasibility and product evolution; organized and tracked delivery using agile practices.

Research, Technological Development and Innovation Intern

03/2020 - 08/2020 | Recife, PE, Brazil | Remote

- Contributed to the development of solutions for molecular diagnosis of SARS-CoV-2 within research, technological development and innovation activities.

Hospital das Clínicas, UFPE / EBSERH

Clinical Pathology Intern

10/2021 - 03/2022 | Recife, PE, Brazil

- Worked in sample screening, hematology, biochemistry, urinalysis, microbiology, hormone assays and serology in the clinical analysis laboratory at HC/UFPE.
- Participated in diagnostic testing, result analysis and quality control procedures, gaining practical experience with clinical laboratory workflows and quality requirements.

RESEARCH EXPERIENCE

Federal University of Minas Gerais - Institute of Biological Sciences

Bioinformatics Researcher | PhD

08/2024 - present | Belo Horizonte, MG, Brazil | On-site

- Develop computational models using deep learning, genomics and structural biology to characterize and design antimicrobial peptides.
- Implement pipelines for sequence processing, molecular-property calculation, structure generation and assessment, and candidate prioritization.
- Build peptide and membrane models and run molecular dynamics simulations to investigate interactions, stability and conformational behavior.
- Analyze how sequence and structural changes affect peptide stability and functionality, supporting the selection of candidates against antimicrobial resistance.
- Develop and integrate code, tools and reproducible workflows that turn genomic and structural data into testable hypotheses and research decisions.

Federal University of Pernambuco (UFPE)

Bioinformatics Researcher | Master's research - Department of Genetics

08/2022 - 07/2024 | Recife, PE, Brazil | On-site

- **Developed AMPIdentifier**, a software tool based on machine learning for identifying antimicrobial peptides in genomic sequences.
- Built the tool's technical workflow, including curation and preparation of large datasets, physicochemical descriptor extraction, classifier training and comparison, and result export.
- Developed a web application and Python package to make the analysis reproducible and accessible to other users.
- Mined and integrated omics data to investigate sequences, molecular signatures, gene expression and functional characteristics of peptides.
- **Computationally identified and characterized five *Ricinus communis* defensins** by combining genomics, transcriptomics, molecular modeling and biological interpretation to support candidate prioritization.

Undergraduate Research Fellow | Plant Genetics and Biotechnology Laboratory

01/2020 - 04/2022 | Recife, PE, Brazil | Hybrid

- Conducted bioinformatics research on the characterization and optimization of bioinspired antimicrobial peptides through sequence and molecular-property analyses.
- Investigated eIF4E family sequences in *Vigna* species and their relationship with defense mechanisms against pathogens, connecting data analysis with functional interpretation.

Undergraduate Research Fellow | Keizo Asami Immunopathology Laboratory (LIKA)

05/2015 - 12/2016 | Recife, PE, Brazil | On-site

- Conducted molecular biology, human genetics and bioinformatics activities in the Molecular Biology sector.
- Investigated ancestry markers and phenotype prediction in forensic genetics, collaborating with the Forensic Genetics Institute of Pernambuco's scientific police.

Oswaldo Cruz Foundation - Aggeu Magalhães Institute

Undergraduate Research Fellow

11/2016 - 12/2019 | Recife, PE, Brazil | On-site

- **Worked in structural biology and theoretical and computational chemistry**, focusing on molecular modeling and protein engineering for diagnostic and vaccine applications.
- **Built and analyzed molecular models** to investigate relationships between protein structure, interactions and function within research conducted in the Department of Virology and Experimental Therapy.
- **Integrated computational analysis and biological interpretation** to characterize molecular systems and inform protein engineering proposals.

SOFTWARE AND BIOINFORMATICS PROJECTS

AMPIdentifier | Antimicrobial peptide prediction

Python, scikit-learn, machine learning, CLI

github.com/madsondeluna/AMPIdentifier

- Developed a tool integrating FASTA sequence input, physicochemical descriptor extraction, classification and result export.
- Made the tool available as a web application and Python package, with installation and usage documentation.
- Software registration with INPI: BR 51 2025 005859-4.

GenVar Dashboard | Human gene and variant exploration

Python, FastAPI, React, Redis, Docker, pytest

github.com/madsondeluna/genvar

- Developed a web application for integrated queries of genes, variants, functional annotations, population frequencies and protein structures.
- Integrated Ensembl, gnomAD, ClinVar, AlphaFold and UniProt, as well as scores provided by MyVariant.info.
- Implemented a front-end, API and cache architecture, containerization, and service and integration tests.
- Developed the project as part of the Software Engineering MBA at USP.

BILBO | Lipid membrane model construction

Python, CLI, system preparation for GROMACS

github.com/madsondeluna/bilbo

- Developed a tool for constructing lipid bilayers and positioning proteins or peptides.
- Generated initial structures, parameters and reports for molecular dynamics simulation preparation.
- Organized the project with documentation and automated tests.

GETVar | Variant processing and annotation

Python, Flask, Snakemake, Bootstrap

github.com/madsondeluna/getvar

- Developed an application for VCF file processing and genetic variant annotation.
- Integrated workflows with sources including dbSNP, ClinVar and Ensembl.

Plant eIF4E Atlas | Biological data integration and exploration

Python, SQL, ETL, APIs and web visualization

github.com/madsondeluna/eif4e-atlas

- Developed a data engineering project to integrate and explore information on plant eIF4E proteins.
- Structured an ETL pipeline and data model for taxonomic and functional queries.
- Developed the project during the Data Science and Analytics specialization at PUC-Rio.

EDUCATION

PhD in Bioinformatics

Federal University of Minas Gerais - UFMG

2024 to present.

MBA in Software Engineering

University of São Paulo - USP

In progress. Expected completion: **December 2026**.

Postgraduate Specialization in Data Science and Analytics

Pontifical Catholic University of Rio de Janeiro - PUC-Rio

Completed in 2026.

Master's degree in Genetics and Molecular Biology

Federal University of Pernambuco - UFPE

Completed in 2024. Research focused on bioinformatics.

Bachelor's degree in Biomedical Sciences

Federal University of Pernambuco - UFPE

Completed in 2022.

Technical Diploma in Software Development

Escola Técnica Estadual de Pernambuco - ETEPE

Completed in 2013.

SELECTED CERTIFICATIONS AND ADDITIONAL TRAINING

- **Microsoft Azure AI Essentials: Workloads and Machine Learning** - 2025.
- **Generative AI** - MIT Professional Education, 2025.
- **Requirements Engineering and Agile Product Management** - PUC-Rio, 2025.
- **Agile Project Management Professional Certificate** - Atlassian, 2025.
- **Career Essentials in GitHub Professional Certificate** - 2024.
- **Advanced Gemini for Developers** - 2024.
- **Slurm Workload Manager** - Laboratório Nacional de Computação Científica, 2022.
- **Multidimensional Arrays and Their Traversal in Python** - CodeSignal, 2022.
- **Nextflow Training** - Seqera.

COMMUNITY AND TEACHING

Seqera - Leadership Program

Student Ambassador (Nextflow Ambassador)

01/2026 - present | Brazil | Hybrid

- Advocate for scalable, reproducible and open-source computational workflows across the bioinformatics and data science communities.
- Share pipeline development best practices and guide researchers in integrating Nextflow and nf-core methodologies into their analysis workflows.
- Collaborate with Seqera developers in public GitHub repositories, contributing to the development and improvement of the company's software products.
- Test tools and workflows across different use cases, identifying unexpected behavior, bugs and opportunities to improve the solutions.
- Create and document GitHub issues with evidence and suggestions that support bug fixes and product evolution.
- Participate in the global scientific community through technical advocacy, knowledge sharing and collaboration in the Nextflow ecosystem. Contributed code, tests and documentation in a merged improvement to Seqera's nf-metro: example contribution.

Invited teaching and educational material development

- PyMOL teaching material for the Basic Bioinformatics Course - 2026.
- Molecular Docking & Dynamics: Breathing Motion into Life's Building Blocks - first and second editions, 2025.
- Molecular Modeling with Machine Learning Techniques - first and second editions, 2025.
- Bioinformatics: A Theoretical-Practical Approach - 2024.
- Introduction to Bioinformatics: From DNA to Proteins - Catholic University of Pernambuco, 2022.

Teaching Assistant | Federal University of Pernambuco

- Human Genetics - 2017 to 2018.
- Molecular Tools Applied to Clinical Diagnosis - 2017 to 2018.

SELECTED PUBLICATIONS

- **de Luna-Aragão, M. A. et al.** Unveiling Three Functionally Diverse Isoforms of eIF4E in Cowpea Through a Multi-Omics Approach. *Agronomy*, 2026. DOI: 10.3390/agronomy16070766.
- **de Andrade, F. A.; de Luna-Aragão, M. A. et al.** Deciphering Cowpea Resistance to Potyvirus: Assessment of eIF4E Gene Mutations and Their Impact on the eIF4E-VPg Protein Interaction. *Viruses*, 2025. DOI: 10.3390/v17081050.
- **Ferraz, M. V. F. et al.** Association strength of E6 to E6AP/p53 complex correlates with HPV-mediated oncogenesis risk. *Biopolymers*, 2022. DOI: 10.1002/bip.23524.

Complete scientific output available on ORCID and Google Scholar.

AWARDS AND HONORS

- Next Generation Bioinformatician (NGB) - 21st International Conference of the Brazilian Association for Bioinformatics and Computational Biology (X-Meeting 2025).
- Travel Grant Award - School on Biological Physics and Biomolecular Simulations in the Machine Learning Era, ICTP-SAIFR.
- Honorable Mention - Human and Forensic Genetics, CNPq/UFPE.
- Young Geneticist Award of the Northeast - XXI Northeast Genetics Meeting (ENGNE).
- Honorable Mention - Postgraduate Genetics Journey, UFPE.
- Best Poster Award - XIII Journey of the Genetics and Molecular Biology Program.
- Travel Grant Award - AI for Protein Design (AI4PD), The Protein Society.
- Certificate of Excellence in Peer Reviewing - Elsevier.
- Highest Admission Score - Ranked first overall among all accepted applicants to the Master's Program in Genetics and Molecular Biology at UFPE.
- Highest Admission Score - Ranked first overall among all accepted applicants to the PhD Program in Bioinformatics at UFMG.
- Honorable Mention - Pernambuco Chemistry Olympiad (OPEQ), National Chemistry Olympiad Program and Pernambuco Science Space.

LANGUAGES

Portuguese: native.

English: advanced.